

Some definitions

set is finite sets implemented as Lists.

Subst := *set* (*Var* * *exp*)

Equation := *set* ((*exp* * *exp*) + (*sexp* * *sexp*))

Problem := *set Equation*

Tuple := *Subst* * *Problem*

⟨**⟩ : *exp* → *Subst* → *exp*

⟨**⟩ : *sexp* → *Subst* → *sexp*

⟨**⟩ : *Problem* → *Subst* → *Problem*

_ @ *_* : *Subst* → *Var* → *exp*

dom : *Subst* → *set Var*

smatch : *Tuple* → *Tuple* → *Prop*

smatch steps can be classified into four

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1. Reflexive steps

$$smatch_refl : (s =_? s) \in P \rightarrow smatch\ (S, P)\ (S, P \setminus (s =_? s))$$

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$$\text{smatch_refl} : (s =_? s) \in P \rightarrow \text{smatch } (S, P) \ (S, P \setminus (s =_? s))$$

2. Congruence steps

$$\begin{aligned} \text{smatch_app_congr} : (s \ t =_? s' \ t') \in P \rightarrow \\ \text{smatch } (S, P) \ (S, P + (s =_? s') + (t =_? t') \setminus (s \ t =_? s' \ t')) \end{aligned}$$

smatch steps can be classified into four

1. Reflexive steps

$$\textit{smatch_refl} : (s =_{\tau} s) \in P \rightarrow \textit{smatch} (S, P) (S, P \setminus (s =_{\tau} s))$$

2. Congruence steps

$$\begin{aligned} \textit{smatch_app_congr} : (s \ t =_{\tau} s' \ t') \in P \rightarrow \\ \textit{smatch} (S, P) (S, P + (s =_{\tau} s') + (t =_{\tau} t') \setminus (s \ t =_{\tau} s' \ t')) \end{aligned}$$

3. σ -min steps

$$\begin{aligned} \textit{smatch_subst_comp} : (s[\sigma \circ \tau] =_{\tau} s'[\rho]) \in P \rightarrow \\ \textit{smatch} (S, P) (S, P + (s[\sigma][\tau] =_{\tau} s'[\rho]) \setminus (s[\sigma \circ \tau] =_{\tau} s'[\rho])) \end{aligned}$$

4. Instantiation case

$$\begin{aligned} \text{smatch_inst} : (s =_? X) \in P \rightarrow \\ X \notin \text{lhvars_of } P \rightarrow \\ S' =_{\sigma_{\min}} S \gg \{(X, s)\} \rightarrow \\ \text{smatch } (S, P) (S', (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle) \end{aligned}$$

Validity

$$\text{valid_tuple } (S, P) := \underline{(\text{dom } S) \cap (\text{vars_of } P) = \phi}$$

Matching solution

$$\text{match_sol } SI \ (S, P) := \frac{(s =_? t) \in P \rightarrow s =_{\sigma_{min}} t \langle SI \rangle \wedge \underline{\exists S', S \gg S' =_{\sigma_{min}} SI}}{}$$

Preservation theorem

$$\begin{aligned} \text{match_pres} : & \text{valid_tuple } T \rightarrow \\ & (\text{dom } SI) \cap (\text{lhvars_of } P) = \phi \rightarrow \\ & \text{smatch } T \ T' \rightarrow \\ & \text{match_sol } SI \ T' \rightarrow \\ & \text{match_sol } SI \ T \end{aligned}$$

Validity and bound variable preservation

$valid_pres : smatch\ T\ T' \rightarrow valid_tuple\ T \rightarrow valid_tuple\ T'$

$lhvars_pres : smatch\ T\ T' \rightarrow (dom\ SI) \cap (lhvars_of\ \pi_2\ T) = \phi \rightarrow (dom\ SI) \cap (lhvars_of\ \pi_2\ T') = \phi$

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smatch is sound

$match_pres : valid_tuple\ T \rightarrow$
 $(dom\ SI) \cap (lhvars_of\ P) = \phi \rightarrow$
 $star\ smatch\ T\ T' \rightarrow$
 $match_sol\ SI\ T' \rightarrow$
 $match_sol\ SI\ T$

Preservation proof proceeds by case analysis on *smatch*

Inst case

$H_0 : s =_? X \in P$

$H_1 : (\text{dom } S) \cap \text{vars_of } P = \phi$

$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \phi$

$H_3 : X \notin \text{lhvars_of } P$

$H_4 : \text{match_sol } SI (S \gg \{(X, s)\}, (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle)$

$\text{match_sol } SI (S, P)$

Inst case (Unfolded *match_sol*)

$$H_0 : s =_? X \in P$$

$$H_1 : (\text{dom } S) \cap \text{vars_of } P = \phi$$

$$H_2 : (\text{dom } Sl) \cap \text{lhvars_of } P = \phi$$

$$H_3 : X \notin \text{lhvars_of } P$$

$$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s =_{\sigma_{min}} t \langle Sl \rangle$$

$$H_5 : \exists S', Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$$

$$\forall s_0, t_0, (s_0 =_? t_0) \in P \rightarrow s_0 =_{\sigma_{min}} t_0 \langle Sl \rangle \quad \wedge$$

$$\exists S', Sl =_{\sigma_{min}} S \gg S'$$

Inst case (Unfolded *match_sol*)

$$\begin{aligned}
H_0 &: s =_? X \in P \\
H_1 &: (\text{dom } S) \cap \text{vars_of } P = \phi \\
H_2 &: (\text{dom } Sl) \cap \text{lhvars_of } P = \phi \\
H_3 &: X \notin \text{lhvars_of } P \\
H_4 &: \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s =_{\sigma_{\min}} t \langle Sl \rangle \\
H_5 &: \exists S', Sl =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'
\end{aligned}$$

$$\begin{aligned}
&\forall s_0, t_0, (s_0 =_? t_0) \in P \rightarrow s_0 =_{\sigma_{\min}} t_0 \langle Sl \rangle \quad \wedge \\
&\exists S', Sl =_{\sigma_{\min}} S \gg S'
\end{aligned}$$

Second conjunct is solvable associativity property of substitution composition.

Inst case continued..

$$H_0 : s =_? X \in P$$

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$$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \phi$$

$$H_3 : X \notin \text{lhvars_of } P$$

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$$H_5 : SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

$$s_0, t_0 : \text{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

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Inst case continued..

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$$H_6 : (s_0 =_? t_0) \in P$$

$$s_0 =_{\sigma_{min}} t_0 \langle SI \rangle$$

Case analysis : $\{s_0 =_? t_0 = s =_? X\} + \{s_0 =_? t_0 \neq s =_? X\}$

Case 1: $s_0 =_? t_0 = s =_? X$

$H_1 : (dom\ S) \cap vars_of\ P = \phi$

$H_2 : (dom\ Sl) \cap lhvars_of\ P = \phi$

$H_3 : X \notin lhvars_of\ P$

$H_4 : \forall\ s'\ t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{min}} t' \langle Sl \rangle$

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$s =_{\sigma_{min}} X \langle Sl \rangle$

$s =_{\sigma_{min}} Sl @ X$ is not immediate

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1. $Y \in (vars_of\ s) \rightarrow Y \notin dom\ Sl$, from H_2 and H_6

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$s =_{\sigma_{\min}} X \langle Sl \rangle$

$s =_{\sigma_{\min}} Sl @ X$ is not immediate

1. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } Sl$, from H_2 and H_6
2. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } (S \gg \{(X, s)\})$, from H_1 and H_3 .

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2. $Y \in (vars_of\ s) \rightarrow Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3 .
3. $(S \gg \{(X, s)\}) @ X = s$, from H_1 and H_6

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2. $Y \in (vars_of\ s) \rightarrow Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3 .
3. $(S \gg \{(X, s)\}) @ X = s$, from H_1 and H_6
4. $((S \gg \{(X, s)\}) \gg S') @ X = ((S \gg \{(X, s)\}) @ X) \langle S' \rangle = s \langle S' \rangle$, from 3 and lookup property

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4. $((S \gg \{(X, s)\}) \gg S') @ X = ((S \gg \{(X, s)\}) @ X) \langle S' \rangle = s \langle S' \rangle$, from 3 and lookup property
5. $Y \in (vars_of\ s) \rightarrow S' @ Y =_{\sigma_{min}} Y$, from a substitution property

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5. $Y \in (vars_of\ s) \rightarrow S' @ Y =_{\sigma_{min}} Y$, from a substitution property
6. $s =_{\sigma_{min}} s \langle S' \rangle$, from 5

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2. $Y \in (vars_of\ s) \rightarrow Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3 .
3. $(S \gg \{(X, s)\}) @ X = s$, from H_1 and H_6
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5. $Y \in (vars_of\ s) \rightarrow S' @ Y =_{\sigma_{min}} Y$, from a substitution property
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Property mentioned in 5:

$$\forall X\ S\ S'\ Sl, X \notin dom\ S \rightarrow X \notin Sl \rightarrow S \gg S' =_{\sigma_{min}} Sl \rightarrow S' @ X =_{\sigma_{min}} X$$

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in \text{vars_of } t_0$

$H_0 : s =_? X \in P$

$H_1 : (\text{dom } S) \cap \text{vars_of } P = \phi$

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$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{min}} t' \langle SI \rangle$

$H_5 : SI =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$

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$s_0 =_{\sigma_{min}} t_0 \langle SI \rangle$

1. We can show $s_0 =_? t_0 \langle \{(X, s)\} \rangle$ $\in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$.

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2. 1 in H_4 gives us $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle SI \rangle$

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$s_0, t_0 : \text{exp}$

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$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$

1. We can show $\underline{s_0 =_? t_0 \langle \{(X, s)\} \rangle} \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle SI \rangle$
3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \gg SI \rangle$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } SI$, from H_2 and H_6

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3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \gg SI \rangle$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } SI$, from H_2 and H_6
5. $SI @ X =_{\sigma_{\min}} s$, same as in case 1.

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$H_6 : (s_0 =_? t_0) \in P$

$s_0 =_{\sigma_{\min}} t_0 \langle Sl \rangle$

1. We can show $\underline{s_0 =_? t_0 \langle \{(X, s)\} \rangle} \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle Sl \rangle$
3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \gg Sl$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } Sl$, from H_2 and H_6
5. $Sl @ X =_{\sigma_{\min}} s$, same as in case 1.
6. $Sl =_{\sigma_{\min}} \{(X, s)\} \gg Sl$, from 4 and 5.

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in \text{vars_of } t_0$

$H_0 : s =_? X \in P$

$H_1 : (\text{dom } S) \cap \text{vars_of } P = \phi$

$H_2 : (\text{dom } Sl) \cap \text{lhvars_of } P = \phi$

$H_3 : X \notin \text{lhvars_of } P$

$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{\min}} t' \langle Sl \rangle$

$H_5 : Sl =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$

$s_0, t_0 : \text{exp}$

$H_6 : (s_0 =_? t_0) \in P$

$s_0 =_{\sigma_{\min}} t_0 \langle Sl \rangle$

1. We can show $\underline{s_0 =_? t_0 \langle \{(X, s)\} \rangle} \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle Sl \rangle$
3. $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \gg Sl \rangle$, due to substitution composition property.
4. $Y \in (\text{vars_of } s) \rightarrow Y \notin \text{dom } Sl$, from H_2 and H_6
5. $Sl @ X =_{\sigma_{\min}} s$, same as in case 1.
6. $Sl =_{\sigma_{\min}} \{(X, s)\} \gg Sl$, from 4 and 5.
7. $t_0 \langle Sl \rangle =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \gg Sl \rangle$

Case 2.2: $s_0 =? t_0 \neq s =? X$, when $X \notin \text{vars_of } t_0$

$H_0 : s =? X \in P$

$H_1 : (\text{dom } S) \cap \text{vars_of } P = \phi$

$H_2 : (\text{dom } SI) \cap \text{lhvars_of } P = \phi$

$H_3 : X \notin \text{lhvars_of } P$

$H_4 : \forall s' t', (s' =? t') \in (P \setminus (s =? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{\min}} t' \langle SI \rangle$

$H_5 : SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$

$s_0, t_0 : \text{exp}$

$H_6 : (s_0 =? t_0) \in P$

$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$

1. Apply H_4 .
2. $(s_0 =? t_0) \in (P \setminus (s, X)) \langle \{(X, s)\} \rangle$ is provable by knowing H_6 , $X \notin \text{vars_of } t_0$, and $s_0 \neq s$ and $t_0 \neq X$.